

Beneficiaries at the Margin: Voter Turnout and State Education Budgets

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October 8, 2026

Abstract

Political support for education spending depends heavily on electoral participation. This paper examines two opposing mechanisms linking voter turnout to state education budgets. Broader participation might amplify demand by increasing the beneficiary share of the electorate, thereby raising political support for spending. Alternatively, it could dilute interest-group influence if organized beneficiaries wield disproportionate power during low-turnout elections but lose relative leverage as the voting pool expands. To evaluate these dynamics, I construct a state-year measure of “adult exposure” as the share of voting-age adults living with school-age children. In a Two-Way Fixed Effects model covering 2012-2022, the interaction is negative across all specifications. At average exposure, higher turnout is associated with a smaller education budget share. While the estimates are imprecise, the results favor interest-group dilution and cast doubt on meaningful demand amplification. Ultimately, the fiscal effect of electoral participation depends entirely on the specific interests it brings into the political process.

JEL Codes: D72, H75, I22

Keywords: median voter, voter turnout, electorate composition, education spending, public choice

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1 Introduction

What determines who governs? There are several viable answers to this question, but voter turnout is undoubtedly one of the most reasonable answers. Changes in voter turnout reorganize and subsequently determine the composition of citizens politicians must satisfy. Despite being an important determinant, expanding the electorate does not push public spending in a consistent direction. Thus, it is incredibly difficult to determine the net effect on electoral outcomes based on a change in turnout.

Before Australia adopted compulsory voting, turnout was heavily skewed toward wealthier citizens. The subsequent reform raised participation by 24 percentage points, simultaneously increasing the Labor Party's vote share and national pension spending (Fowler 2013). In contrast, when Austrian states adopted similar compulsory voting laws, turnout rose by roughly ten percentage points with little discernible effect on the level or composition of government spending (Hoffman, León, and Lombardi 2017). From what the evidence shows us, a larger electorate is as likely to bring more program beneficiaries into the political process as it is to mobilize taxpayers with greater preferences for fiscal restraint. Ultimately, the fiscal effect depends on which of these two groups shows up at the polls.

The public choice literature posits two competing accounts of this divergence. Demand amplification, rooted in median voter models (Downs 1957; Meltzer and Richard 1981), suggests that broader participation shifts the decisive voter toward new beneficiaries who demand more spending. On the other hand, collective action theory (Olson 1965; Hardin 1982) points toward interest-group dilution, where organized constituencies hold disproportionate sway during low-turnout elections, with their relative influence weakening as the voting pool expands. In this paper, I ask what determines the prevailing outcome. I focus on state education budgets during ordinary federal election cycles and how responsive a beneficiary constituency's turnout is relative to the rest of the electorate.

Public education provides a natural setting for this study. Households with school-

age children maintain a recurring interest in educational services, while the tax burden is spread across a much broader population. I capture the potential political reach of this constituency using a measure of “adult exposure”, representing the share of voting-age adults living with a child aged 5 to 17. Interacting this measure with lagged voter turnout in a fixed-effects model of state expenditure from 2012 to 2022 reveals a trade-off among competing claims on state revenue.

If broader participation amplifies demand, higher turnout in high-exposure states should bring more interested voters to the polls, yielding a positive interaction. However, if it dilutes organized influence, the interaction should be negative. Across all specifications, the interaction is negative, and higher turnout is associated with a smaller education budget share at average exposure. Though the estimates are noisy, they lean heavily toward dilution and are difficult to reconcile with a theory of demand amplification.

The existing evidence falls into two camps depending on the source of turnout variation. Findings that support amplification typically rely on reforms that lowered participation costs for specific beneficiary groups (G. Miller 2008; Cascio and Washington 2014; Fujiwara 2015; Bertocchi et al. 2020). In contrast, dilution findings emerge primarily from election timing shifts (Anzia 2011, 2014) and general cost reductions that target no particular demographic (Hodler, Luechinger, and Stutzer 2015). While a few studies suggest a conditional framework, evidence drawn from ordinary, untargeted election-cycle variation remains thin (Dynes, Hartney, and Hayes 2021; Klien, Melki, and Pickering 2021).

This paper makes two primary contributions. First, it identifies relative turnout responsiveness as the common mathematical condition linking demand amplification and interest-group dilution. Rather than treating them as competing theories, this framework positions them as two outcomes of a single participation process. This allows for a unified interpretation of the targeted-reform and election-timing literature, while further developing the citizen-driven side of median voter endogeneity. Although Dunne, Reed, and Wilbanks (1997) and Bierbrauer, Tsyvinski, and Werquin (2022) emphasize how

elites shape participation through election rules and strategic demobilization, my account highlights how organic differences in citizens' participation responses can independently shift the effective median.

Second, the paper examines this framework in a distinct empirical setting. Instead of relying on institutional reforms that mechanically alter voting costs for specific sub-populations, I utilize ordinary federal election-cycle variation. This untargeted setting provides crucial evidence on how beneficiary influence changes when participation fluctuates naturally through the electoral process rather than through structural interventions.

Section 2 develops the theoretical mechanisms and the turnout-responsiveness condition that separates them. Section 3 describes the data and the construction of the adult exposure measure. Section 4 presents the empirical strategy. Section 5 reports the estimates and sensitivity checks. Section 6 discusses the implications for collective provision, and Section 7 concludes.

2 Participation and Political Influence

Politicians respond to the active electorate rather than the broader eligible population. As turnout changes, the resulting shifts in electoral size and composition alter the preferences of the median voter. For public finance, the net fiscal effect depends entirely on the preferences of the marginal citizens entering or leaving the voting pool.

2.1 The Median Voter as a Product of Participation

The median voter model tells us that political competition draws policy toward the center of the electoral distribution (Downs 1957). Applying this logic to public finance, Meltzer and Richard (1981) demonstrate how the location of that decisive voter shapes redistribution and the overall size of government.

Because turnout rates vary substantially across socioeconomic groups, the median of the participating electorate often diverges from that of the eligible population (Lijphart

1997; Highton and Wolfinger 2001; Franko, Kelly, and Witko 2016; Kaplow and Kominers 2024). Elected officials face strong incentives to serve these reliable voters, given that abstainers impose little to no electoral penalty when policy moves against their interests (Peltzman 1976; Griffin and Newman 2005; Larcinese 2007).

Shifts in participation can relocate this effective median voter. On the strategic side, politicians can deliberately reshape the active electorate through institutional rules, election dates, or targeted policy proposals (Dunne, Reed, and Wilbanks 1997). Rather than treating low turnout as an accident, parties may craft platforms specifically to depress opposing turnout, making the non-voting public a calculated equilibrium outcome (Bierbrauer, Tsyvinski, and Werquin 2022).

Alongside elite maneuvering, a compositional channel opens up whenever broader economic conditions or close races pull different demographic groups to the polls. For example, Klien, Melki, and Pickering (2021) show that municipal education spending in Austria responds to turnout differently depending on the local elderly population. A generic rise in turnout could mean one thing if it mobilizes low-income voters or parents, and something entirely different if it turns out suburban taxpayers or public employees.

2.2 Participation, Concentrated Benefits, and Fiscal Constraint

Voter participation links the median voter model to collective action problems. Public programs typically deliver concentrated benefits funded through diffuse costs. Beneficiaries possess strong incentives to organize because their policy stakes are highly visible (Olson 1965), utilizing formal networks and recurring institutional contact to coordinate political pressure (Hardin 1982). Meanwhile, taxpayers face weaker mobilization incentives since the financial burden imposed on any single individual remains small.

Elected officials must balance the political rewards of satisfying these concentrated beneficiaries against the electoral risks of alienating tax-sensitive voters (Peltzman 1976). In low-turnout environments, organized beneficiaries constitute a disproportionately large share of the active electorate, minimizing the political cost of taxing inactive citizens.

However, as turnout rises, diffuse interests and unorganized taxpayers cast a longer electoral shadow, increasing the political cost of directing resources toward a single, concentrated constituency (Denzau and Munger 1986).

Whether an influx of voters strengthens an interested constituency or expands the broader electorate that constrains it depends almost exclusively on whether marginal voters stand to benefit or simply foot the bill. Historical expansions that lowered voting costs for specific demographic groups reliably shifted spending in their favor, whether from public health investments following women’s suffrage (G. Miller 2008) or targeted transfers under the Voting Rights Act (Cascio and Washington 2014). In contrast, when voting costs fall across the board, the resulting surge in participation tends to dilute concentrated interests and restrain public spending (Hodler, Luechinger, and Stutzer 2015).

2.3 Education Spending and Concentrated Interests

Public education fits this dynamic well, pairing broad social spillovers with concentrated immediate benefits. A well-educated population supports local labor markets and property values (Fischel 2006; Hilber and Mayer 2009), but households with enrolled children capture the day-to-day returns. Since school policy shapes their daily routines and their children’s future, these parents have a strong personal stake and face very low costs to monitor local school decisions.

Research consistently demonstrates that household composition and age structure shape political support for education expenditures (Rubinfeld 1977; Poterba 1997; Harris, Evans, and Schwab 2001; Brunner and Balsdon 2004). Furthermore, off-cycle election timing often depresses turnout in local school governance, enabling organized employee and education interests to secure favorable outcomes (Anzia 2011, 2014; Kogan, Lavertu, and Peskowitz 2018; Hartney 2022). Alternatively, moving municipal elections on-cycle produces an electorate that is structurally more representative and less dominated by narrow interest groups (Hajnal, Kogan, and Markarian 2022).

The share of parents varies considerably across states and over time, dictating the potential political reach of education-linked constituencies. C. Miller (1996) establishes an empirical foundation for treating constituency size as a determinant of spending, finding that education budgets increase alongside the number of parents in the electorate. The more important question is whether broader turnout amplifies that specific constituency’s political influence or dilutes it.

The net effect depends on how school-connected turnout responds relative to the rest of the electorate. Let e denote adult exposure, the share of voting-age adults connected to public schooling, and let t_B and t_N denote the turnout rates of school-connected adults and other adults. Overall turnout is

$$T = e t_B + (1 - e) t_N, \quad (1)$$

and the school-connected share of the active electorate is

$$s = \frac{e t_B}{e t_B + (1 - e) t_N}. \quad (2)$$

Consider any change in the electoral environment that raises overall turnout, and define the responsiveness of each group’s participation as $\eta_B = \partial \ln t_B / \partial \ln T$ and $\eta_N = \partial \ln t_N / \partial \ln T$. Differentiating T gives $s \eta_B + (1 - s) \eta_N = 1$, so the two responsiveness terms are a weighted average of unity. Because $\ln s = \ln e + \ln t_B - \ln T$,

$$\frac{\partial \ln s}{\partial \ln T} = \eta_B - 1 = (1 - s) (\eta_B - \eta_N). \quad (3)$$

With $0 < s < 1$, the sign of the compositional shift is the sign of $\eta_B - \eta_N$. Broader participation raises the school-connected share of the electorate when that group’s turnout is more responsive than other adults’ turnout. The school-connected share is lower when their turnout is less responsive. Therefore, Demand amplification and interest-group dilution are not competing theoretical models. They are simply the two cases of equation (3), distinguished by the responsiveness of the beneficiary constituency’s turnout relative

to everyone else’s.

Evidence regarding the costs of voting suggests which case is more likely when turnout variation targets no specific demographic. Cantoni, Pons, and Schafer (2025) find that voting frictions matter most when the stakes of an election are low or diffuse. Because school-connected adults have a recurring interest in public education, their turnout should be relatively inelastic, responding less to general turnout swings than that of citizens without school-age children.

3 Data

I assemble a state-year panel linking electoral participation, household composition, and state education spending across 49 states for 2012, 2014, 2016, 2018, and 2022. Nebraska is excluded because its nonpartisan unicameral legislature has no party control. I exclude 2020 because pandemic disruptions led the Census Bureau to withhold standard American Community Survey (ACS) one-year estimates and release microdata only with experimental weights. These five federal election cycles provide a consistent match between fiscal outcomes, ACS demographics, and election turnout. Massachusetts in 2014 and Alaska in 2022 are excluded because of reporting anomalies in their spending data, leaving 243 observations in the estimating sample.

3.1 Education Spending and Turnout

The dependent variable comes from the U.S. Census Bureau’s Annual Survey of State Government Finances. I measure public K-12 education spending as a share of total state expenditure. This budget share captures how state leaders prioritize education against competing public uses of revenue, effectively controlling for overall revenue growth and making large and small states analytically comparable.

Turnout is the share of the voting-eligible population that participated in the preceding federal election. I lag turnout by two years to pair each fiscal year with its most

recent presidential or midterm election, preserving the theoretical sequence where citizens vote first, officials write budgets second, and spending choices appear in subsequent fiscal records. I mean-center turnout before estimation. State and year fixed effects absorb time-invariant state characteristics and nationwide shocks, ensuring identification relies exclusively on within-state variation across election cycles.

3.2 Adult Exposure

The primary demographic measure comes from the American Community Survey Public Use Microdata Sample. I define adult exposure as the weighted share of voting-age adults living in households with at least one child between ages 5 and 17. Using person-level microdata, household identifiers determine whether each adult lives with a school-age child, and ACS person weights aggregate these observations to the state-year level. Formally,

$$AdultExposure_{st} = \frac{\sum_{i \in A_{st}} w_{ist} \mathbb{I}(Child_{h(i),st} = 1)}{\sum_{i \in A_{st}} w_{ist}} \quad (4)$$

where A_{st} is the set of voting-age adults in state s and year t , w_{ist} is the individual ACS sampling weight, and $\mathbb{I}(Child_{h(i),st} = 1)$ equals one if adult i resides in a household $h(i)$ with at least one child aged 5 to 17. I mean-center adult exposure before estimation.

Adult exposure measure potential political breadth. Because a single child can link multiple voting-age adults to the public school system without increasing the student count, this measure effectively isolates the voter-side multiplier. It identifies the share of adults for whom education policy holds an immediate household connection without requiring assumptions about their explicit voting behavior.

3.3 Enrollment Burden and Controls

Enrollment burden is public school enrollment relative to total state population, using data from the National Center for Education Statistics Common Core of Data. While

adult exposure maps political reach, enrollment burden captures the direct administrative volume of students the state must educate. The pooled correlation between the two measures is 0.748 and falls to 0.68 within states, confirming they track closely but retain independent variation.

Additional demographic controls from ACS state-level estimates include the bachelor's degree share, the white voting-age population share, and the Gini coefficient to account for changes in educational attainment, racial composition, and income inequality. To control for broader economic and political conditions, I include the annual state average unemployment rate and partisan composition indicators tracking unified Republican or Democratic control over the governorship and legislature.

3.4 Descriptive Statistics

Table 1 presents summary statistics. Education spending averages 16.1 percent of total state expenditure, ranging from 8.3 to 26.4 percent. Because identification relies on within-state variation, interpretation of adult exposure should rely on within-state distribution instead of the overall sample variance. Including both adult exposure and enrollment burden allows me to separate the political dynamics of connected households from the baseline pressures of service demand.

Table 1: Descriptive Statistics

Variable	N	Mean	SD	Min.	Median	Max.
Education spending share	243	0.161	0.034	0.083	0.157	0.264
Turnout _{<i>t</i>-2}	245	0.547	0.124	0.281	0.569	0.792
Adult exposure	245	0.280	0.027	0.206	0.278	0.374
Enrollment burden	245	0.153	0.016	0.118	0.152	0.217
Bachelor’s degree share	245	0.309	0.056	0.186	0.303	0.466
White voting-age population share	245	0.769	0.133	0.242	0.793	0.961
Gini coefficient	245	0.464	0.019	0.408	0.466	0.521
Unemployment rate	245	4.980	1.848	1.900	4.500	11.500
Republican control	245	0.473	0.500	0.000	0.000	1.000
Democratic control	245	0.220	0.415	0.000	0.000	1.000

Notes: Turnout and adult exposure are reported in levels; both are mean-centered prior to estimation. The panel excludes Nebraska. The education spending outcome contains 243 observations following the exclusion of Massachusetts in 2014 and Alaska in 2022; all other variables contain 245 observations.

4 Empirical Strategy

If turnout changes the political influence of education-linked households, its association with education spending should depend on adult exposure. I estimate the following two-way fixed-effects model:

$$\begin{aligned}
 EducationShare_{st} = & \beta Turnout_{s,t-2} + \gamma AdultExposure_{st} \\
 & + \delta(Turnout_{s,t-2} \times AdultExposure_{st}) + X'_{st}\theta + \alpha_s + \lambda_t + \epsilon_{st}
 \end{aligned} \tag{5}$$

$EducationShare_{st}$ is the share of state expenditure devoted to elementary and secondary education. $Turnout_{s,t-2}$ is mean-centered voting-eligible turnout in the preceding federal election. $AdultExposure_{st}$ is the mean-centered share of voting-age adults living with a school-age child. Both variables are centered at their means in the 245-observation analysis panel, 0.547 for turnout and 0.280 for adult exposure. X_{st} is the vector of controls discussed previously. State fixed effects, α_s , absorb persistent institutional and fiscal differences across states, while year fixed effects, λ_t , absorb shocks common to all states in a given year. A positive δ aligns with demand amplification, while a negative δ

aligns with interest-group dilution.

4.1 Interpreting the Interaction

The interaction coefficient δ captures how the turnout slope changes with adult exposure:

$$\frac{\partial EducationShare_{st}}{\partial Turnout_{s,t-2}} = \beta + \delta AdultExposure_{st} \quad (6)$$

Centering both variables places each constituent term at a reference point inside the data. β is the marginal effect of turnout at the sample average of adult exposure, and γ is the association between adult exposure and education spending at the sample average turnout rate. Following Brambor, Clark, and Golder (2006), I evaluate the marginal effect at the 25th, 50th, and 75th percentiles of the within-state exposure distribution to properly match the source of identification.

4.2 Specifications

Table 2 builds the specification over four columns. Column 1 includes turnout, adult exposure, their interaction, and the fixed effects. Column 2 adds enrollment burden to test whether the relationship survives after accounting for the volume of students. Column 3 incorporates the demographic controls, and column 4 completes the preferred specification by adding unemployment and party control.

4.3 Inference and Sensitivity

Standard errors are clustered by state to account for observations within a state remaining correlated over time. With 49 clusters, conventional robust inference is generally workable, though finite-sample performance remains a consideration (Cameron and Miller 2015). I subsequently check the preferred specification by adding state-specific linear trends to address slow-moving fiscal trajectories, dropping 2022 to exclude pandemic-era federal aid anomalies, and redefining adult exposure to include children aged 0 to 17.

5 Results

5.1 Main Estimates

Table 2 reports the primary estimates. The interaction between turnout and adult exposure is negative in every column, ranging from -0.386 to -0.465. In the preferred specification (column 4), the interaction is -0.422 with a clustered standard error of 0.234, significant at the 10 percent level. Incorporating enrollment burden, demographics, and partisan controls moves the estimate only marginally, indicating the negative interaction is not an artifact of service demand or sweeping partisan shifts.

Turnout itself exhibits a negative association with education budget share at average adult exposure. Reaching the 10 percent significance level in column 4, a 10-percentage-point increase in turnout corresponds to an education budget share roughly 0.25 percentage points lower in an average-exposure state. If demand amplification held, higher turnout would not shrink education’s share on average.

The adult exposure coefficient, evaluated at average turnout, remains small and imprecise across all specifications. Because the data do not identify a stable independent association between constituency size and the education budget share, the results rely heavily on the conditional relationship.

5.2 Marginal Associations

Table 3 details the marginal association of turnout across low, typical, and high levels of adult exposure. Higher turnout negatively correlates with education budget share throughout the exposure distribution. A 10-percentage-point increase in turnout corresponds to a 0.22-percentage-point decline at the 25th percentile of exposure, dropping further to a 0.27-percentage-point decline at the 75th percentile.

The difference across this interquartile range is -0.52 percentage points, carrying a 95 percent confidence interval from -1.11 to 0.06. Because this interval crosses zero, the estimates structurally cannot prove that higher exposure actively deepens the negative

Table 2: Main Specification: Turnout, Adult Exposure, and Education Spending

	(1)	(2)	(3)	(4)
Turnout _{<i>t</i>-2}	-0.023 (0.018)	-0.014 (0.016)	-0.023 (0.014)	-0.025* (0.014)
Adult Exposure	0.135 (0.158)	0.046 (0.150)	0.057 (0.136)	0.065 (0.132)
Turnout _{<i>t</i>-2} × Adult Exposure	-0.436 (0.271)	-0.465* (0.268)	-0.386* (0.230)	-0.422* (0.234)
Enrollment Burden		0.851* (0.457)	1.00** (0.475)	1.05** (0.482)
Bachelor's degree share			0.260* (0.149)	0.245 (0.153)
White voting-age population share			0.026 (0.031)	0.033 (0.032)
Gini coefficient			-0.145 (0.204)	-0.128 (0.210)
Unemployment rate				-0.001 (0.001)
Republican control				-0.004 (0.003)
Democratic control				0.003 (0.002)
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	243	243	243	243
Within R^2	0.034	0.063	0.093	0.116

Notes: Column 4 is the preferred specification. Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

relationship. However, the upper bound strictly limits any theoretical positive shift in turnout’s marginal association to a mere 0.06 percentage points, drastically constraining any argument for demand amplification.

Table 3: Marginal Association of Turnout Across Adult Exposure

	Estimate	95% Confidence Interval	
25th percentile of adult exposure	-2.19	-5.12	0.74
50th percentile of adult exposure	-2.48	-5.30	0.34
75th percentile of adult exposure	-2.72	-5.48	0.04
75th–25th percentile difference	-0.52	-1.11	0.06

Notes: Estimates come from the preferred specification in column 4 of Table 2. Entries in the first three rows report the estimated marginal association between turnout and education spending, in percentage points, evaluated at the 25th, 50th, and 75th percentiles of the within-state adult exposure distribution. The 75th–25th percentile difference reports the change in turnout’s marginal association across the within-state interquartile range of adult exposure. Confidence intervals are based on standard errors clustered by state.

5.3 Sensitivity

Table 5 in the appendix outlines the three sensitivity checks. The interaction remains negative across all alternative constructions: -0.467 with state-specific trends, -0.448 when dropping the 2022 cycle entirely, and -0.306 when expanding adult exposure to include children up to age 17. While none achieve standalone statistical significance, the sign and approximate magnitude remain highly stable.

6 Discussion

My central result is that a larger education-linked constituency does not make turnout more favorable to education spending. The interaction is negative across specifications, and higher turnout is associated with a smaller education budget share at average exposure. While the estimates are imprecise, they lean toward dilution. Broader participation appears to weaken the political advantage associated with a larger directly interested constituency.

6.1 Limits of Voter-Driven Expansion

C. Miller (1996)’s finding that a larger parent constituency is associated with greater education spending creates a natural expectation that broader turnout should reinforce that relationship, but the estimates in this study do not support that expectation. In states where more voting-age adults live with school-age children, higher turnout does not translate into larger education budget shares.

One explanation is that school-connected adults already participate at relatively high rates through PTAs, school networks, and other recurring institutions (Hardin 1982). If so, increases in general turnout may bring non-parent voters into the electorate faster than parents. Constituency size is still important for baseline political influence, but its marginal advantage shrinks as participation broadens. Returning to the condition established in Section 2, the turnout of school-connected adults should be less responsive than other adults’ turnout and broader participation should be diluted rather than amplified.

The result identifies a practical limit to demographic reach as a measure of political power. Population share matters, but it must be weighed alongside other factors such as habitual participation and the specific composition of marginal turnout.

6.2 Interest Group Dilution

Collective action theory explains why dilution is the likely regime for education finance (Olson 1965; Hardin 1982). Because school spending provides visible benefits to parents and school employees, these groups possess strong incentives to organize and monitor policy. The costs are spread across a much broader taxpayer base, leaving individual taxpayers with weaker incentives to follow school spending closely.

Low turnout magnifies this asymmetry because organized beneficiaries make up a larger share of the active electorate. In the political support framework of Peltzman (1976), policymakers face a high electoral cost from disappointing concentrated voters and a comparatively low cost from taxing citizens who rarely participate. The school-

board setting analyzed by Anzia (2011, 2014) makes the mechanics of this intuitive.

In a low-turnout election, organized parent networks and school employees can account for a large share of votes cast, while diffuse taxpayers remain underrepresented. When participation rises, those organized interests remain active, but they are joined by voters with weaker ties to public schooling and stronger competing fiscal claims. Following Denzau and Munger (1986), diffuse interests become more politically relevant simply because they enter the voting pool.

In this case, higher turnout acts as a structural counterweight. The new voters do not need to actively oppose public education; they only need to place relatively greater weight on taxes, infrastructure, health programs, public safety, or other state priorities. The concentrated education constituency then competes within a broader set of politically relevant claims. This is consistent with pressure-group models of competition among organized interests (Becker 1983). It's also consistent with Fleck (1999), where political benefits flow toward groups with greater electoral value.

Therefore, the negative interaction is consistent with a change in relative political influence. Education-linked households remain an important constituency, but their leverage can decline as the voting pool expands. This represents the compositional form of median voter endogeneity. While Dunne, Reed, and Wilbanks (1997) emphasize strategic changes to election rules and timing, the relevant median here shifts because the identity and preferences of participating voters organically change with turnout.

This helps explain why median voter models can perform unevenly across empirical studies. Researchers might measure the demographic median of the eligible or total population, while policy responds to the median of the active electorate. When participation differs systematically across groups, those two medians can diverge significantly.

6.3 Competing Fiscal Claims

Because the dependent variable is education's share of total state expenditure, the result is best interpreted as a budget-allocation effect. A lower share does not necessarily mean

nominal or per-pupil spending falls. Rather, it means education receives a smaller portion of the state budget relative to competing uses of revenue. This means voters can broadly support public education while still preferring more spending on health care, transportation, pensions, public safety, or tax relief. Essentially, broader participation may constrain education’s budget share without requiring outright hostility toward schools.

Adult exposure should be interpreted similarly, as it identifies the reach of an immediate household connection to education. Exposed adults differ vastly in income, ideology, school type, home-ownership, satisfaction with public schools, and preferences over efficiency or school choice. The political content of the measure is salience.

6.4 Limits of the Dilution Interpretation

I cannot identify the specific citizens who enter the electorate when turnout rises or determine whether they possess weaker ties to public education, stronger preferences for fiscal restraint, or greater demand for competing programs. In particular, the estimates of this study do not reveal whether school-connected adults’ turnout is mathematically more or less responsive than other adults’, which is the quantity both explanations depend on. If parties set platforms partly to demobilize the other side’s supporters (Bierbrauer, Tsyvinski, and Werquin 2022), relative turnout responsiveness is itself an equilibrium outcome, and the estimated interaction serves as a reduced form of that equilibrium rather than a structural elasticity.

Furthermore, adult exposure does not measure formal organization through parent groups, teachers’ unions, taxpayer associations, campaign contributions, or lobbying. The measure captures parents rather than school employees, who historically form much of the organized core in the election-timing literature. Evidence from school board elections shows how far the education electorate can diverge from the education constituency. Kogan, Lavertu, and Peskowitz (2021) document considerable demographic differences between the voters who participate in school board elections and the students the boards serve. Thus, the dilution interpretation does not require parents to be a stable organized

bloc. However, it does show that voters entering at higher turnout have weaker ties to public schooling than those already voting. Additionally, while the preferred specification holds unemployment and party control fixed, factors like election competitiveness, candidate quality, and issue salience may affect both turnout and budget priorities simultaneously.

My bounded conclusion is that the results lean toward dilution and are difficult to reconcile with meaningful demand amplification. They do not, however, identify the exact channel through which broader participation changes education's budget share.

6.5 Future Research

Further research should focus on the specific voters producing this compositional change. Validated turnout records linked to household structure could show whether school-connected adults are more or less likely to enter the electorate when participation rises. The Current Population Survey Voting and Registration Supplement records self-reported turnout alongside household rosters, and offers a way to compare the turnout of adults living with school-age children against other adults across election years. Voter-file data could also identify how marginal participants differ in age, income, home-ownership, or party registration.

Incorporating explicit measures of organization would help separate demographic exposure from active political mobilization. Union membership, parent-group activity, school-board participation, campaign contributions, and education lobbying could reveal whether concentrated education interests are indeed strongest when general participation is low. Institutional features such as initiative rules, tax and expenditure limits, school-finance centralization, gubernatorial budget authority, and legislative professionalism may also shape whether turnout amplifies demand or dilutes it.

I would also like to extend this research beyond the education setting. Medicaid, pensions, agricultural subsidies, higher education, transportation projects, and occupational regulation are all forms of collective provision that distribute benefits and costs unevenly.

In each case, whether broader participation mobilizes beneficiaries or dilutes their relative influence depends on how responsive that group’s turnout is relative to other voters’ turnout. It also depends on whether the source of turnout variation targets the group.

7 Conclusion

I examine whether turnout changes state education spending through the composition of the electorate, treating demand amplification and dilution as two outcomes of the same political process. Broader participation strengthens a directly interested constituency when that constituency’s turnout responds more than other voters’ turnout, and weakens its relative influence when it responds less.

The estimates lean toward dilution, with the turnout-adult exposure interaction being negative in every specification and significant at the 10 percent level in the preferred specification. Higher turnout is associated with a smaller education budget share at average exposure, though the estimates are imprecise. This pattern holds with state-specific trends, without 2022, and with an alternative definition of adult exposure, and it is difficult to reconcile with meaningful demand amplification.

The main contribution of this paper is identifying the condition that decides between demand amplification and interest group dilution. Whether higher turnout expands or shrinks an interested constituency’s influence depends on how responsive that group’s turnout is relative to everyone else. That condition explains why reforms targeting beneficiaries expand their influence (G. Miller 2008; Cascio and Washington 2014; Fujiwara 2015), while election timing and untargeted cost cuts dilute it (Anzia 2014; Hodler, Luechinger, and Stutzer 2015), and why compulsory voting changed policy in some places but not others. The evidence here adds the untargeted case. In ordinary election cycles, where no reform determines who shows up, state education budgets lean toward dilution.

For state education budgets, the findings suggest that broader participation acts as a political counterweight. Higher turnout widens the electorate beyond directly inter-

ested and organized constituencies, forcing policymakers to accommodate more voters who share the tax burden and weigh education against competing public priorities. Because the analysis does not observe marginal voters or formal organization directly, that mechanism remains provisional. Based on the empirical results, I conclude that a larger education-linked adult population does not make turnout more expansionary in the manner demand amplification predicts.

For public choice, the broader implication is that participation can operate in two directions. It can mobilize beneficiaries, or it can dilute their relative influence by making diffuse interests politically relevant, depending on whose turnout moves. The fiscal effect of turnout therefore depends on the electorate that higher participation creates.

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A Figures

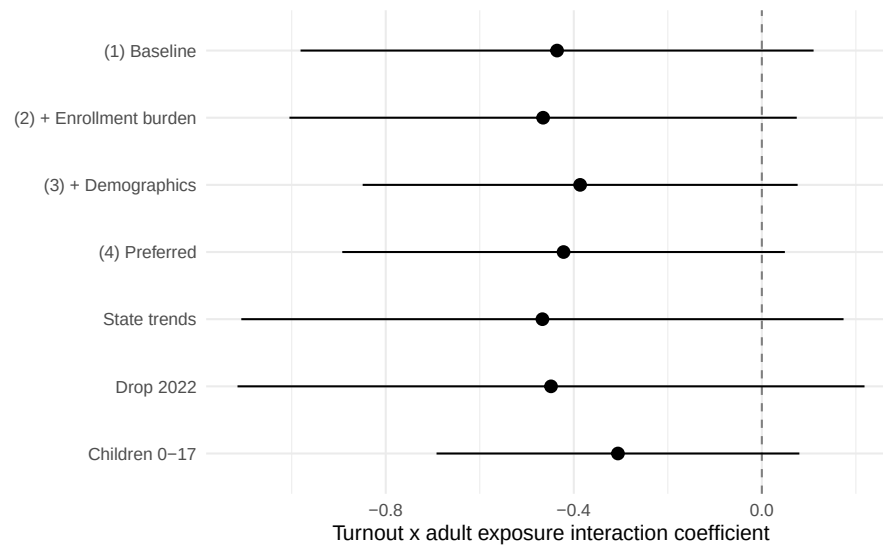


Figure 1: Turnout $\times$ adult exposure interaction across specifications. Points are the estimated interaction coefficient with 95 percent confidence intervals based on standard errors clustered by state. Columns 1 through 4 correspond to Table 2; the remaining three correspond to the sensitivity checks in Table 5. The dashed line marks zero.

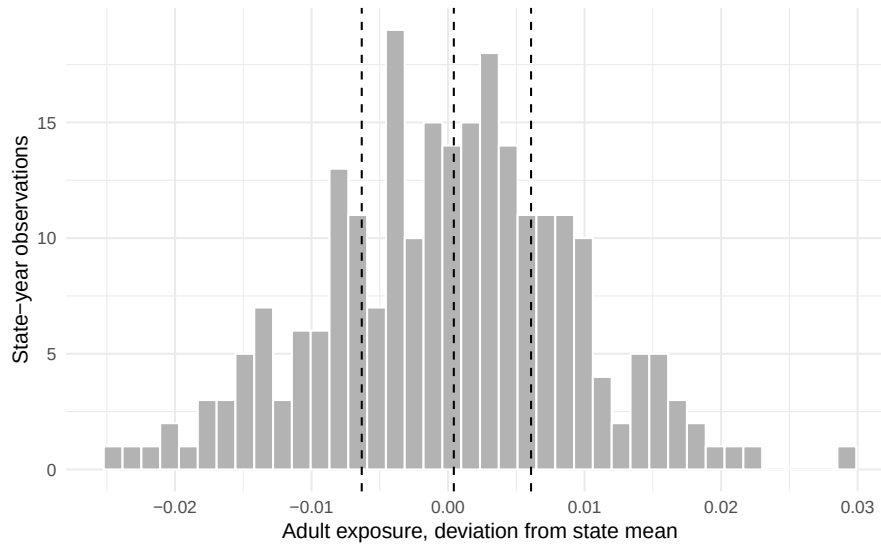


Figure 2: Within-state distribution of adult exposure. Each observation is a state-year deviation from its state mean, the variation that identifies the interaction under state and year fixed effects. Dashed lines mark the 25th, 50th, and 75th percentiles, the points at which Table 3 evaluates the marginal association of turnout.

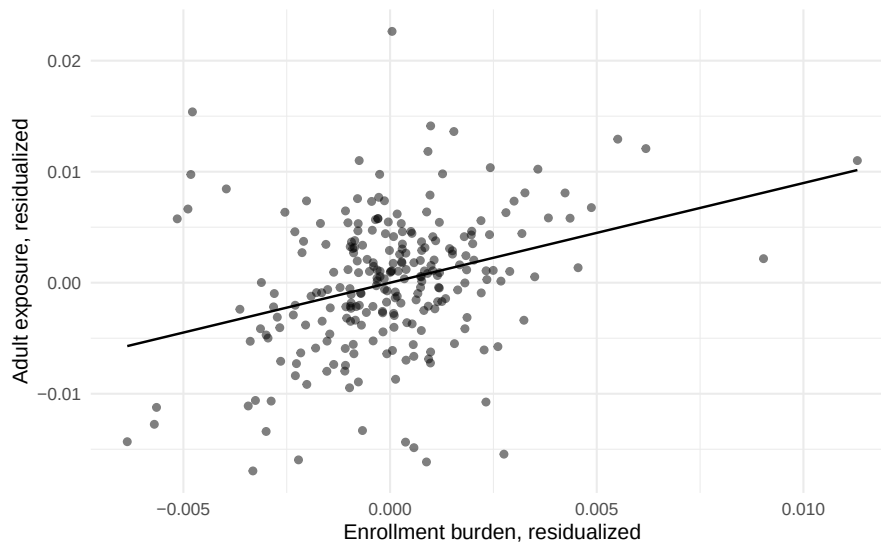


Figure 3: Adult exposure and enrollment burden after absorbing state and year fixed effects. Each point is a state-year residual; the line is the linear fit. The residualized correlation is 0.68, indicating that the two measures move together while adult exposure retains variation independent of service demand.

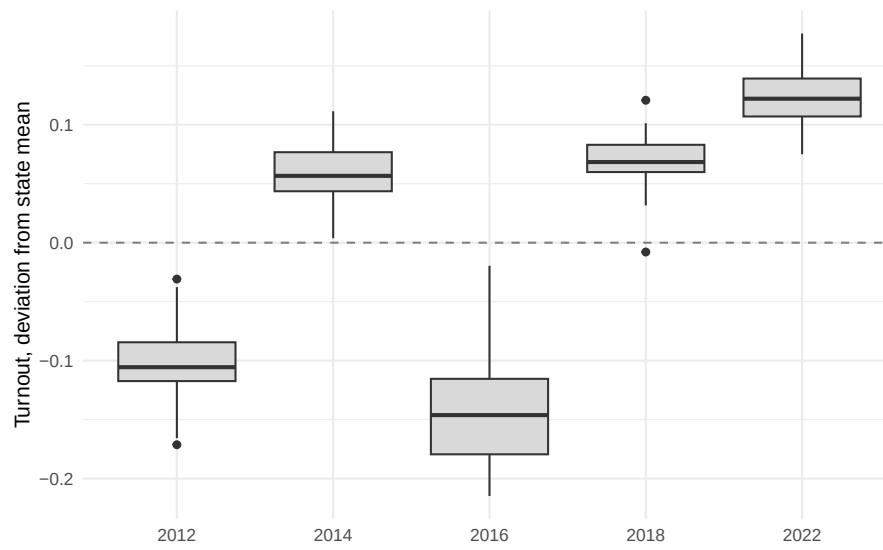


Figure 4: Within-state turnout by election cycle. Boxes show the distribution of state-year deviations from state mean turnout across the five federal elections in the panel. The dashed line marks zero. Presidential and midterm cycles generate variation an order of magnitude larger than the within-state variation in adult exposure shown in Figure A2.

B Variable Definitions and Data Construction

This appendix provides variable details, underlying data sources, and construction methods for the state-year panel. The panel covers 49 states across 2012, 2014, 2016, 2018, and 2022. Nebraska is excluded because its nonpartisan unicameral legislature has no party control. Massachusetts in 2014 and Alaska in 2022 are excluded from regression specifications using state education spending share due to reporting discontinuities in the finance series. The panel begins in 2012 because comparable state government finance measures are broadly unavailable in 2010 and 2011. I exclude 2020 because pandemic-related data collection problems led the Census Bureau to withhold its standard ACS one-year estimates and release the PUMS only with experimental weights that are not directly comparable with standard estimates from other years. These restrictions allow a consistent match between fiscal outcomes, ACS household data, and turnout in the preceding federal election.

Adult exposure is calculated from ACS microdata using individual person weights

$$AdultExposure_{st} = \frac{\sum_{i \in A_{st}} w_{ist} \mathbb{1}(Child_{h(i),st} = 1)}{\sum_{i \in A_{st}} w_{ist}} \quad (7)$$

where A_{st} represents voting-age adults in housing units, w_{ist} is the ACS person weight, and the indicator equals one if adult i resides with a child aged 5 to 17. The alternative measure used in the sensitivity checks replaces the 5 to 17 age range with 0 to 17.

Table 4 outlines all variable definitions and data sources.

Table 4: Variable Definitions and Data Sources

Variable	Definition	Source and Construction
Education spending share	Elementary and secondary education expenditure divided by total state expenditure	U.S. Census Bureau Annual Survey of State Government Finances. Excludes Massachusetts (2014) and Alaska (2022).
Turnout _{$t-2$}	Voting-eligible turnout in the preceding federal election	University of Florida Election Lab. Lagged two years to align election cycles with subsequent state budget allocations. Mean-centered prior to model estimation.
Adult exposure	Share of voting-age adults residing with at least one child aged 5 to 17	ACS Public Use Microdata Sample. Person-weighted aggregation. Mean-centered prior to model estimation.
Adult exposure, ages 0–17	Share of voting-age adults residing with at least one child aged 0 to 17	Constructed as above. Used in the sensitivity checks.
Enrollment burden	Public school enrollment relative to state population	NCES Common Core of Data and ACS state totals.
Demographics	Bachelor’s degree share, white voting-age population share, and Gini coefficient	ACS state-level estimates.
Unemployment rate	Annual average state unemployment rate, in percent	Bureau of Labor Statistics, Local Area Unemployment Statistics, not seasonally adjusted annual averages.
Party control	Indicators for Republican and Democratic control of the governorship and both legislative chambers; divided government is the reference category	NCSL State and Legislative Partisan Composition tables for each panel year. New York and Washington in 2014 are coded Democratic as reported by NCSL.

C Sensitivity Checks

Table 5 reports the preferred specification alongside three alternatives: state-specific linear trends, a sample that drops 2022, and adult exposure redefined to include children aged 0 to 17.

Table 5: Sensitivity of the Preferred Specification

	Preferred (1)	State trends (2)	Drop 2022 (3)	Children 0–17 (4)
Turnout _{<i>t</i>−2}	−0.025* (0.014)	−0.021 (0.018)	−0.031** (0.013)	−0.027* (0.014)
Adult exposure	0.065 (0.132)	0.006 (0.154)	−0.010 (0.099)	0.058 (0.110)
Turnout _{<i>t</i>−2} × Adult exposure	−0.422* (0.234)	−0.467 (0.319)	−0.448 (0.332)	−0.306 (0.192)
Controls	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State-specific linear trends	No	Yes	No	No
Observations	243	243	195	243
Within R^2	0.116	0.094	0.149	0.113

Notes. Dependent variable is education spending as a share of total state expenditure. Every column includes enrollment burden, the bachelor’s degree share, the white voting-age population share, the Gini coefficient, the unemployment rate, and indicators for Republican and Democratic control. In column 4, adult exposure is the share of voting-age adults residing with at least one child aged 0 to 17. Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.